

JOONWON LEE

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Goldman Sachs
Global Banking & Markets
Global Currency and Emerging Markets
New York, NY

Re: Quantitative Strategist, Global Currency and Emerging Markets (GCEM)

Dear GCEM Hiring Team,

I am applying for the Quantitative Strategist position within Goldman Sachs' Global Currency and Emerging Markets team. I am a Statistics Ph.D. candidate at Rutgers University and expect to complete my degree in December 2026. I bring rigorous training in stochastic modeling, machine learning, numerical optimization, and simulation, together with applied experience in quantitative finance, insurance pricing, and algorithmic market making. I am particularly drawn to the desk-strat model because it combines quantitative research, production-oriented coding, and close collaboration with traders on commercially important problems.

At JPMorgan Chase, I developed and validated a Sequential Probability Ratio Test framework for binary risk indicators. I translated a monitoring problem into statistically controlled decision rules and designed simulation-based evaluation of Type I and Type II error, detection delay, and early-decision tradeoffs. Brownian-motion approximations helped characterize stopping-time uncertainty and communicate the model's behavior and limitations. This experience strengthened my ability to turn an ambiguous business question into a precise quantitative framework and evaluate it under realistic operating tradeoffs.

My doctoral research develops scalable approximate-likelihood inference and statistical diagnostics for large, dependent datasets. I built an advection-aware Vecchia approximation for nonseparable Gaussian processes, replacing dense covariance operations with fixed-budget conditioning to fit up to 145,008 satellite observations per day. I implemented the work in Python and PyTorch using GLS profiling, L-BFGS optimization, CPU/GPU execution, and restartable HPC workflows. Although the application is atmospheric data, the underlying work has trained me to calibrate and test models under dependence, missingness, noise, and computational constraints.

I have also applied machine learning to business pricing decisions. At Travelers, I developed an end-to-end LightGBM pricing pipeline across more than 2.46 million property-risk records, benchmarked it against a generalized linear model, evaluated performance and feature attribution, and presented the results as pricing and risk-segmentation recommendations. Separately, I placed in the top 2.39% of participants in the IMC Prosperity Algorithmic Trading Competition by designing a market-making strategy with inventory-aware quoting, dynamic liquidation rules, and benchmark-driven evaluation. That experience reinforced my interest in building quantitative systems that respond to market and risk conditions.

I would welcome the opportunity to bring this combination of statistical depth, practical Python engineering, and model-validation discipline to GCEM. I am eager to deepen my product knowledge across Rates, FX, and Credit while contributing to machine-learning research, market-making optimization, and robust trading tools. Thank you for your consideration.

Sincerely,

Joonwon Lee